

Nudge — Session Handoff Context

Date: March 22, 2026

Product Overview

Nudge is a full-stack AI reading companion. Users save URLs or paste text; Nudge summarizes content via LLMs and presents a weekly digest. Philosophy: "Read fewer things. Think more deeply."

Target: AI Infra PM portfolio project — specifically targeting Fluidstack AI Platform PM and similar roles.

Live URLs:

- Frontend: <https://nudge-repo.vercel.app>
 - Backend: <https://nudge-repo-production.up.railway.app>
 - GitHub: <https://github.com/souvikai/nudge-repo>
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Current Production Stack

Layer	Tech	Platform
Frontend	Next.js 16, TypeScript, Tailwind	Vercel
Backend API	FastAPI, Python 3.11	Railway
Database	PostgreSQL 16	Railway
Worker	Python polling worker (airy-reflection)	Railway
Digest Cron	Python cron job (creative-sparkle)	Railway
AI Strong Tier	Gemini 2.5 Flash	Google AI
AI Mid Tier	DeepSeek deepseek-chat	DeepSeek
AI Budget Tier	DeepSeek deepseek-chat	DeepSeek

Local dev: Docker Compose at `C:\Users\sou90\nudge-repo\` **Branch strategy:** `deploy` branch → production (same as `main`); push to both `deploy` and `deploy:main --force`

Key Files

Backend

- `app/main.py` — FastAPI app, CORS, routers (items, auth, benchmark, analytics)
- `app/api/items.py` — all item endpoints
- `app/api/auth.py` — magic link auth endpoints
- `app/api/benchmark.py` — benchmark endpoint (runs all 3 tiers in parallel)
- `app/api/analytics.py` — analytics/observability endpoint
- `app/auth.py` — JWT session token validation
- `app/models/mvp.py` — SQLAlchemy models
- `app/settings.py` — config (RESEND_API_KEY, JWT_SECRET, FRONTEND_URL, LLM tiers)

- `app/llm/client.py` — MULTI-PROVIDER: DeepSeek + Gemini, provider-agnostic routing
- `app/worker/worker.py` — URL extraction worker
- `app/worker/digest.py` — weekly digest email sender
- `run_digest.py` — entry point for Railway cron (uses `sys.path.insert` to fix module resolution)

Frontend

- `frontend/src/app/page.tsx` — main UI
- `frontend/src/app/benchmark/page.tsx` — model benchmark dashboard (dark theme)
- `frontend/src/app/analytics/page.tsx` — inference observatory dashboard (dark theme)
- `frontend/src/app/login/page.tsx` — magic link login page
- `frontend/src/app/auth/verify/page.tsx` — token verify + redirect
- `frontend/src/lib/itemsApi.ts` — API client with auth token
- `frontend/src/lib/types.ts` — TypeScript types

.env Current State (Local)

```

DATABASE_URL=postgresql+psycopg://postgres:postgres@localhost:5432/nudge?sslmode=disable
LLM_DEFAULT_MODEL_KEY=mid
LLM_MID_PROVIDER=deepseek
LLM_MID_MODEL=deepseek-chat
LLM_MID_API_KEY=sk-b05967d64c524557b3482cdab3825db1
LLM_STRONG_PROVIDER=gemini
LLM_STRONG_MODEL=gemini-2.5-flash
LLM_STRONG_API_KEY=AIzaSyDytb6vTysMt4Ythqs2zpPb7uqU7-jIbu4
LLM_BUDGET_PROVIDER=deepseek
LLM_BUDGET_MODEL=deepseek-chat
LLM_BUDGET_API_KEY=sk-b05967d64c524557b3482cdab3825db1
RESEND_API_KEY=re_ZX4CJvWk_PbyaitgwSivaQnuLWxJLWpKd
JWT_SECRET=nudge-super-secret-jwt-key-change-in-production-2026
FRONTEND_URL=http://localhost:3000

```

Railway Services

nudge-repo (Backend API)

- All `.env` vars above PLUS:
- `FRONTEND_URL`=<https://nudge-repo.vercel.app>
- `DATABASE_URL`=`postgresql+psycopg://postgres:...`[@postgres.railway.internal](https://railway.com):5432/railway
- `LLM_STRONG_PROVIDER`=gemini, `LLM_STRONG_MODEL`=gemini-2.5-flash, `LLM_STRONG_API_KEY`=`AIzaSyDytb6vTysMt4Ythqs2zpPb7uqU7-jIbu4`
- Pre-deploy command: `alembic upgrade head`

airy-reflection (URL Extraction Worker)

- Same env vars as `nudge-repo`
- Runs `python -m app.worker` (polls for queued items continuously)

creative-sparkle (Digest Cron)

- Same env vars as `nudge-repo`
- Start command: `python /app/run_digest.py`

- Schedule: `0 9 * * 0` (every Sunday 9am UTC = 2:30pm IST)
- IMPORTANT: Must use `run_digest.py` at repo root — `python -m app.worker.digest` fails due to Railway module path issue

API Endpoints (All Working)

Method	Path	Purpose
POST	/items	Save URL or text
GET	/items	List items
GET	/items/{id}	Get item detail
PATCH	/items/{id}/text	Submit fallback text
GET	/items/{id}/summary	Get cached summary
POST	/items/{id}/summary	Generate new summary
POST	/auth/request-login	Send magic link email
GET	/auth/verify	Verify token, return session
POST	/benchmark/{item_id}	Run all 3 tiers in parallel
GET	/analytics	Inference observability data
GET	/health	Health check

Alembic Migrations Applied

1. `20260125_0001` — create mvp schema
2. `20260221_0002` — add item_summaries + summary_attempts
3. `20260320_0003` — add auth (email on users + magic_tokens table)

What Was Completed This Session

1. ☒ Benchmark page (`/benchmark`) — dark mission control dashboard, 3-column model cards, latency/cost bars, badges
2. ☒ Inference Observatory (`/analytics`) — per-tier p50/p95 latency, success rates, 14-day SVG bar chart, call share bars
3. ☒ Analytics backend (`GET /analytics`) — PostgreSQL percentile_cont with Python fallback
4. ☒ Nav links on main page — "Model Benchmark →" and "Inference Observatory →"
5. ☒ Weekly digest email (`app/worker/digest.py`) — sends via Resend to all users with summaries
6. ☒ Railway cron job (creative-sparkle) — runs Sundays 9am UTC via `run_digest.py`
7. ☒ Gemini env vars added to Railway nudge-repo service

Benchmark Key Findings (Real Data)

- Gemini (strong): ~3.5s, ~\$0.0052 — FASTEST but 7x more expensive
- DeepSeek mid: ~6-7s, ~\$0.0007 — good balance

- DeepSeek budget: ~8s, ~\$0.0007 — cheapest, slowest
- Default production tier: mid (DeepSeek)

What Needs To Be Done Next (Priority Order)

1. Stress Test Suite (HIGHEST PRIORITY — for interviews)

Build a script that runs 10-15 golden articles through the benchmark to populate the Observatory with real data. Need diverse content types: Wikipedia, HackerNews, arXiv, public Substack, personal blogs, Reddit. Add a simple quality score (word overlap or 1-5 human rating). Build a cost projection at 100 WAU doing 5 summaries/week.

2. Jina Reader Integration

Integrate `https://r.jina.ai/` as a fallback before "needs user text" — handles many JS-rendered sites that the current extractor fails on. Prepend to URL before fetch attempt.

3. Custom Domain + Email Deliverability

Buy a domain (nudge.so or getnudge.app), add to Resend, set up SPF/DKIM DNS records. Change FROM_ADDRESS in `digest.py` from `onboarding@resend.dev` to `digest@yourdomain.com`. Currently emails land in spam.

4. Update Session Handoff Doc

Keep this doc updated after each session.

Frontend Pages Summary

Page	URL	Theme	Purpose
Main	/	Yellow (Nudge brand)	Save articles, weekly digest
Login	/login	Yellow	Magic link login
Benchmark	/benchmark	Dark (mission control)	Compare model tiers
Observatory	/analytics	Dark (mission control)	Inference observability

AI Infra PM Portfolio Story

Resume bullet: "Built Nudge, a full-stack LLM-powered knowledge capture system with multi-provider model routing (Gemini + DeepSeek), live benchmarking dashboard comparing latency/cost/quality across tiers, inference observability pipeline with p50/p95 metrics, and automated weekly digest via Railway cron — deployed on Railway + Vercel"

Target roles: Fluidstack AI Platform PM (\$180-250k), similar AI infra PM roles

Technical Gotchas To Remember

- Windows: `str_replace` unreliable, use Cursor editor for file changes
- PowerShell: backtick template literals get stripped, use Cursor for JS/TS files
- Docker Compose: must restart containers after `.env` changes
- Gemini 2.5 Flash: needs `thinkingBudget: 0` and `maxOutputTokens: 2048`

- Gemini API: uses `x-goog-api-key` header
- Deploy: always push to both `deploy` branch AND `deploy:main --force`
- Railway cron: use `run_digest.py` at repo root (NOT `python -m app.worker.digest`) — module path issue
- Tailwind v4: arbitrary color classes don't work reliably in new pages — use inline styles instead
- Tailwind v4: page-specific dark background requires `document.body.style.background` override in `useEffect`